

ECON 101 Lecture Notes: Demand and Supply

These notes accompany weeks three and four of ECON 101. They were written for the Rocky public evaluation corpus and are released under the Apache-2.0 license.

Demand

Economists use the word demand to describe the relationship between the price of a good and the quantity of that good that buyers are willing and able to purchase at each price. Quantity demanded is the specific amount bought at one particular price.

The law of demand states that, other things being equal, a higher price leads to a lower quantity demanded and a lower price leads to a higher quantity demanded. The phrase *ceteris paribus*, Latin for other things being equal, reminds us that every other factor that affects demand is held constant while we vary the price.

A demand schedule is a table that shows the quantity demanded at each price. A demand curve plots the same information as a graph with price on the vertical axis and quantity on the horizontal axis, and it slopes downward from left to right.

Supply

Supply describes the relationship between price and the quantity that producers are willing to sell. The law of supply states that a higher price leads to a higher quantity supplied, because higher prices make production more profitable. A supply curve therefore slopes upward from left to right.

Equilibrium: The Gasoline Example

Equilibrium is the price at which quantity demanded equals quantity supplied, so there is no pressure for the price to change. In our gasoline example the market reaches equilibrium at a price of \$1.40 per gallon, where both quantity demanded and quantity supplied are 600 million gallons.

Price per gallon	Quantity demanded (millions)	Quantity supplied (millions)
\$1.00	800	500
\$1.40	600	600
\$1.80	500	680

At any price above equilibrium, quantity supplied exceeds quantity demanded and the market has excess supply, also called a surplus. At any price below equilibrium, quantity demanded exceeds quantity supplied, which creates excess demand, also called a shortage.

What Shifts Demand

A change in price moves us along a demand curve, while a change in any other factor shifts the whole curve. The main factors that shift demand are income, tastes and preferences, the prices of related goods, the size and composition of the population, and expectations about future prices.

A normal good is one for which demand rises when income rises. An inferior good is one for which demand falls when income rises, for example bus travel for people who switch to driving once they can afford a car.

Substitutes are goods that can replace each other, such as coffee and tea, so a higher price for one raises demand for the other. Complements are goods used together, such as golf clubs and golf balls, so a higher price for one lowers demand for the other.

What Shifts Supply

The main factors that shift supply are the costs of production, including wages and raw materials, changes in technology, natural conditions such as weather, and government policies such as taxes, subsidies and regulations. A new technology that lowers production costs shifts the supply curve to the right.

Price Ceilings and Price Floors

A price ceiling is a legal maximum on the price at which a good can be sold. Rent control is a common example of a price ceiling in housing markets. When a price ceiling is set below the equilibrium price, it creates a shortage, because quantity demanded exceeds quantity supplied at the controlled price.

A price floor is a legal minimum on the price at which a good can be sold. The minimum wage is the best known example of a price floor in a labor market. When a price floor is set above the equilibrium price, it creates a surplus, because quantity supplied exceeds quantity demanded.

Exam Reminder

The midterm exam covers weeks one through four and takes place on October 15 in Hall B. Bring a calculator; no notes are allowed.